EEG WORKSHOP: Investigating electroencephalographic signals through different analytic paradigms

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Objectives

A three-module intensive workshop tailored to the datasets, experimental protocols, and research priorities of the EEG laboratories at *Istanbul Medipol University*, providing hands-on training in the analysis of oscillatory and arrhythmic activities, avalanche-like events and EEG-informed biological brain age. Each module integrates a dedicated segment on emerging research frontiers, fostering the identification of concrete opportunities for scientific collaboration.

Program

- *Oscillations in EEG signals.*
  Filtering. Hilbert transform, Circular statistics. **Analytic examples.**
  Coherence measures (Spectral coherence, PLV, wPLV, PPC, PLI, wPLI).
  Cross frequency couplings (PAC, PFC, PPC). Cross-Frequency Directionality (CFD). Mechanistic interpretations (Theta-Gamma code, Gamma-leading Theta interactions). **Coding examples.**
  Phase Clustering (PC). **Coding examples.**
  Spectral harmonicity (TLI). **Coding examples.**
  Statistical analysis: Nonparametric permutation testing. **Coding examples.**
  Ideas for collaboration: CFC directionality. CFC and functional networks. CFC and working memory. CFC and neuromodulation. Assessing cortical excitability using EEG and tES. Transcranial Temporal Interference Stimulation (tTIS).

- *Broadband arrhythmic activity in EEG signals.*
  Neuronal avalanches and criticality hypothesis. Aperiodic and scale-free activities. Long-range temporal correlations. Detrended fluctuation analysis scaling exponent and spectral aperiodic exponent.
  Linking oscillations and transient salient events: Spectral group delay consistency. **Analytic and Coding examples.**
  Avalanche-like events in EEG as a biomarker of neurological disorders. **Coding examples for Parkinson's disease.**
  Ultradian dynamics of salient events.
  Ideas for collaboration: Ultradian dynamics of salient events in brain disorders. Avalanche-like events and neuromodulation.

- *Brain Aging Models.*
  Normative modeling based on neuroimaging.
  Brain aging charts: Brain feature ~ Chronological age + covariates
  Brain Clock Models: Biological brain age ~ Brain feature(s)
  Biological age vs. chronological age.
  Functional, phase-based, group delay-based and O-info connectivities.
  Regression model for building the clock. Methodological caveats.
  Non-invasive biophysical neuromodulation to reduce functional brain aging. **Coding examples.**
  Ideas for collaboration: Brain clocks based on arrhythmic activity. Brain clocks and neuromodulation.

Resources

https://github.com/damian-dellavale/EEG_workshop

Schedule

3 August 2026: Oscillations in EEG signals
10:00 - 11:30: Filtering. Hilbert transform. Coherence measures.
11:30 - 11:45: Coffee Break.
11:45 - 13:15: Coherence measures.
13:15 - 14:15: Lunch Break.
14:15 - 15:45: Cross-Frequency Coupling (CFC). Cross-Frequency Directionality (CFD). Mechanistic interpretations. Phase Clustering (PC). Spectral harmonicity (TLI).
15:45 - 16:00: Break.
16:00 - 17:00: Statistical analysis: Nonparametric permutation testing.

4 August 2026: Broadband arrhythmic activity in EEG signals
10:00 - 11:30: Neuronal avalanches and criticality hypothesis. Aperiodic and scale-free activities. Long-range temporal correlations. Detrended fluctuation analysis scaling exponent and spectral aperiodic exponent.
11:30 - 11:45: Coffee Break.
11:45 - 13:15: Linking oscillations and transient salient events: Spectral group delay consistency.
13:15 - 14:15: Lunch Break.
14:15 - 15:45: Avalanche-like events in EEG as a biomarker of neurological disorders.
15:45 - 16:00: Break.
16:00 - 17:00: Ultradian dynamics of salient events.

5 August 2026: Brain Aging Models
10:00 - 11:30: Normative modeling based on neuroimaging. Brain aging charts.
11:30 - 11:45: Coffee Break.
11:45 - 13:15: Brain Clock Models. Biological age vs. chronological age.
13:15 - 14:15: Lunch Break.
14:15 - 15:45: Connectivity measures. Regression model for building the clock. Methodological caveats.
15:45 - 16:00: Break.
16:00 - 17:00: Non-invasive biophysical neuromodulation to reduce functional brain aging.

6 and 7 August 2026:
10:00 - 13:15: Q&A. Revisiting specific topics of interest. Hands-on coding. Collaboration ideas.
14:15 - 17:00: Q&A. Revisiting specific topics of interest. Hands-on coding. Collaboration ideas.